CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of covalent bonds in butane C_4H_{10} is: 1

(a) 13 **(b)** 10 **(c)** 14 **(d)** 12

2. CH_3CH_2OH on heating with excess conc. H_2SO_4 at 443 K gives: 1

(a) $CH_2=CH_2$ **(b)** CH_3CH_3 **(c)** CH_3CHO **(d)** CH_3COOH

3. Which of these is NOT a property of carbon? 1

(a) catenation **(b)** tetravalency

(c) forms multiple bonds **(d)** forms ionic bonds with chlorine

4. The number of covalent bonds in propane C_3H_8 is: 1

(a) 9 **(b)** 11 **(c)** 8 **(d)** 10

5. CH_3CH_2OH with alkaline $KMnO_4$ and heat gives: 1

(a) $CH_3COOC_2H_5$ **(b)** CH_3CH_3 **(c)** $CH_2=CH_2$ **(d)** CH_3COOH

6. Carbon cannot form C_4^+ because: 1

(a) it has 2 valence electrons

(b) it has 8 valence electrons

(c) it is a gas

(d) high energy is needed to remove 4 electrons

7. The number of shared electron pairs in ethene (C_2H_4) is: 1

(a) 8 **(b)** 4 **(c)** 6 **(d)** 5

8. Which of the following will decolourise bromine water? 1

(a) C_2H_6 **(b)** C_2H_4 **(c)** C_3H_8 **(d)** CH_4

9. Carbon cannot form C_4^- because: 1

(a) it has 8 electrons

(b) it is heavy

(c) it is metal

(d) 6 protons cannot hold 10 electrons easily

10. The number of shared electron pairs in ethyne (C_2H_2) is: 1

(a) 3 **(b)** 5 **(c)** 6 **(d)** 4

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does carbon form covalent compounds only and not ionic ones? 2

12. How will you test for unsaturation in a hydrocarbon? 2

13. Write the reaction of methanoic acid with sodium hydroxide. 2

14. Differentiate between a covalent and an ionic compound in melting point and conductivity. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Calculate the number of covalent bonds in 3
(a) hexane C_6H_{14} and
(b) $C_{10}H_{22}$.

16. Compare the properties of soap and detergent. 3

17. What is a covalent bond? Show formation of Cl_2 and O_2. 3

OR

Explain why diamond and graphite have different properties though both are carbon. 3

18. Explain the formation of micelles in soap solution. 3

OR

Describe how ethanoic acid differs from hydrochloric acid in reactivity. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain tetravalency, catenation and covalent bonding of carbon with three reasons for the large number of carbon compounds. 5

OR

Describe the substitution and addition reactions and the conditions with equations; also classify the reaction of ethene with bromine. 5

20. Compare ethane, ethene and ethyne in formula, bonds, combustion and reactions. 5

OR

Name the first six members of alkane series with number of C-H and C-C bonds in each. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND